

Mini AI Knowledge Assistant - Sample Document

Section 1: What is Retrieval-Augmented Generation (RAG)?

Retrieval-Augmented Generation, commonly known as RAG, is an artificial intelligence framework designed to improve the quality of large language model responses by grounding the model on external knowledge sources. While standard language models are trained on large corpuses of data, their knowledge is static and cut off at their training date. Furthermore, models can generate factually incorrect information, a phenomenon known as hallucination.

RAG addresses these challenges by retrieving authoritative facts from an external knowledge base before generating a response. This allows the model to reference timely and private data, substantially reducing the likelihood of hallucinations.

Section 2: The Core RAG Pipeline

The architecture of a typical RAG system consists of three fundamental stages:

1. Ingestion: Documents such as PDFs or text files are parsed, cleaned, and split into smaller manageable segments called text chunks.
2. Indexing: Each chunk is passed through an embedding model to produce a high-dimensional vector representation. These vectors are stored in a vector database such as FAISS or Chroma.
3. Retrieval and Generation: When a user poses a question, the query is converted into an embedding. The vector database performs a cosine similarity search to find the most relevant chunks. Finally, the retrieved chunks and the user query are assembled into a prompt and submitted to an LLM.

Section 3: Chunking Strategies and Grounded Answering

Chunking is the process of breaking long documents into cohesive, smaller units. Two primary parameters govern chunking: chunk size and chunk overlap. Chunk size determines the maximum length of each fragment, while chunk overlap ensures that context spanning chunk boundaries is not lost.

To guarantee grounded answers, the system prompt must explicitly instruct the LLM to answer only using the retrieved facts. If the information is missing from the context, the assistant must state that it cannot find the answer rather than speculating.

